Fusion services

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IBM Fusion Service provides scalable, integrated storage and data management solutions optimized for hybrid cloud and AI workloads on Red Hat® OpenShift®.

Global Data Platform service

The Global Data Platform (GDP) in IBM Fusion is a unified data management solution that simplifies access, sharing, and management of data across diverse environments. It offers scalable storage optimized for hybrid cloud and AI workloads, ensuring seamless data availability and mobility across multiple locations. It is a software-defined file and object storage that enables organizations to build a global data platform for artificial intelligence (AI), high-performance computing (HPC), advanced analytics, and other demanding workloads.

The Global Data Platform storage type provides the following features:
- File storage
- High availability via capacity-efficient erasure coding
- Metro and regional disaster recovery
- CSI snapshot support with built-in application consistency
- Encryption at rest
- Ability to mount file systems hosted by remote IBM Storage Scale clusters.

Fusion Data Foundation

You can use Fusion Data Foundation as storage service in IBM Fusion. It enables you to manage Fusion Data Foundation services, and to deploy and scale storage clusters. The Fusion Data Foundation storage type provides the following features:
- Block, file, and object storage
- High availability through automatic data replication
- Metro and regional disaster recovery
- CSI snapshot support
- Encryption at rest

For more information about Fusion Data Foundation service, see Introduction to Fusion Data Foundation.

Backup & Restore

The Backup & Restore service provides the following features:
- Policy-driven backup of applications that run on Red Hat OpenShift

- Change block detection for Ceph RBD block volumes. It greatly reduces backup times for applications using block mode volumes, including OpenShift Virtualization VMs based on RBD block volumes.
- Orchestration of application consistent online backups through recipes
- Multi-cluster Backup & Restore by using a hub and spoke topology

The Backup & Restore provides application-centric backup and data recovery. To know more about the service and its architecture, see Backup & Restore overview.

IBM Data Cataloging

The IBM Data Cataloging provides data insight for exabyte-scale heterogeneous file, object, backup, and archive storage on premises and in the cloud. The software easily connects to these data sources to rapidly ingest, consolidate, and index metadata for billions of files and objects. For more information about the IBM Data Cataloging service and its architecture, see Data Cataloging.

Content-Aware Storage (CAS)

It leverages natural language processing (NLP) and AI technologies to extract semantic meaning from unstructured data, such as text, charts, graphs, and images. It allows for more efficient data retrieval and integration with AI workflows, particularly in retrieval-augmented generation (RAG) processes. For more information about the CAS service, see Content-Aware Storage (CAS).

- **IBM Fusion services and platform support matrix**
  Support matrix of IBM Fusion services on the different deployment platforms of IBM FusionSupport matrix of IBM Fusion services on the Bare Metal deployment platform of IBM Fusion HCI.
- **Fusion Data Foundation**
  Welcome to the official documentation for IBM Fusion Data Foundation 4.20.
- **Global Data Platform**
- **Backup & Restore**
  The Backup & Restore service protects your workloads with tailored backups by using a hub and spol model. You can restore your data rapidly from local snapshots without having to rely on external storage. Backups can be transferred to external object storage solutions like IBM Cloud, Microsoft Azure, Amazon Web Services, and any S3-compliant object storage. It allows the protection of the IB Fusion Backup & Restore service itself to a S3 bucket.
- **Data Cataloging**
  Companies need the ability to use unstructured data to meet their business priorities.
- **Content-Aware Storage (CAS)**
  The IBM Fusion Content-Aware Storage service is engineered to enhance the value of customers’ AI applications, such as retrieval augmented generation (RAG) for faster time to insights, reduced cost, improved performance, enhanced security, and streamlined operations.
- **Deploying IBM Fusion services using CLI**
  Use the instructions to deploy the IBM Fusion services using the command-line interface (CLI).

Parent topic:

→ Fusion services and capabilities